

Li et al. (2024) reanalysis

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```
library(tidyverse)
library(readxl)
library(janitor)
library(gt)
```

1 Load data

Supplementary data obtained from <https://doi.org/10.1161/CIRCHEARTFAILURE.124.011725> on 20250429. Here we load the protein abundance data from Table S2 and the sample metadata from Table S1.

```
li_2024_abundance = readxl::read_xlsx("data/circhf-2024-011725d-s01.xlsx",
  sheet = "Table S2",
  skip = 2,
  na = "NA") %>%
  dplyr::rename(gene = ...1) # add name for column contain gene names or symbols

li_2024_sample_meta = readxl::read_xlsx("data/circhf-2024-011725d-s01.xlsx",
  sheet = "Table S1",
  skip = 2,
  na = "NA") %>%
  clean_names() %>%
```

```

dplyr::mutate(across(where(is.character), as.factor))

# wrangle into tidy format
all(colnames(select(li_2024_abundance, -gene)) %in% li_2024_sample_meta$sample_id) # check all samples

li_2024_abundance_tidy = li_2024_abundance %>%
  pivot_longer(cols = -gene,
               names_to = "sample_id",
               values_to = "abundance") %>%
  left_join(li_2024_sample_meta, by = "sample_id") %>%
  dplyr::mutate(condition = factor(condition, levels = c("Donor", "DCM", "PPCM"))) # reorder factor levels

```

2 Calculate Percentage of Missing Values

In Figure 3H of the authors paper, there are fewer data points than the stated sample size. Here we calculate the overall average percentage of values that are missing in the proteomics dataset, across all proteins and all samples.

```

li_2024_abundance_tidy %>%
  summarise(n_missing = sum(is.na(abundance)),
            total_expected_values = n(),
            percent_missing = n_missing / total_expected_values * 100) %>%
  gt()

```

n_missing	total_expected_values	percent_missing
13703	125280	10.9379

3 SBSPON

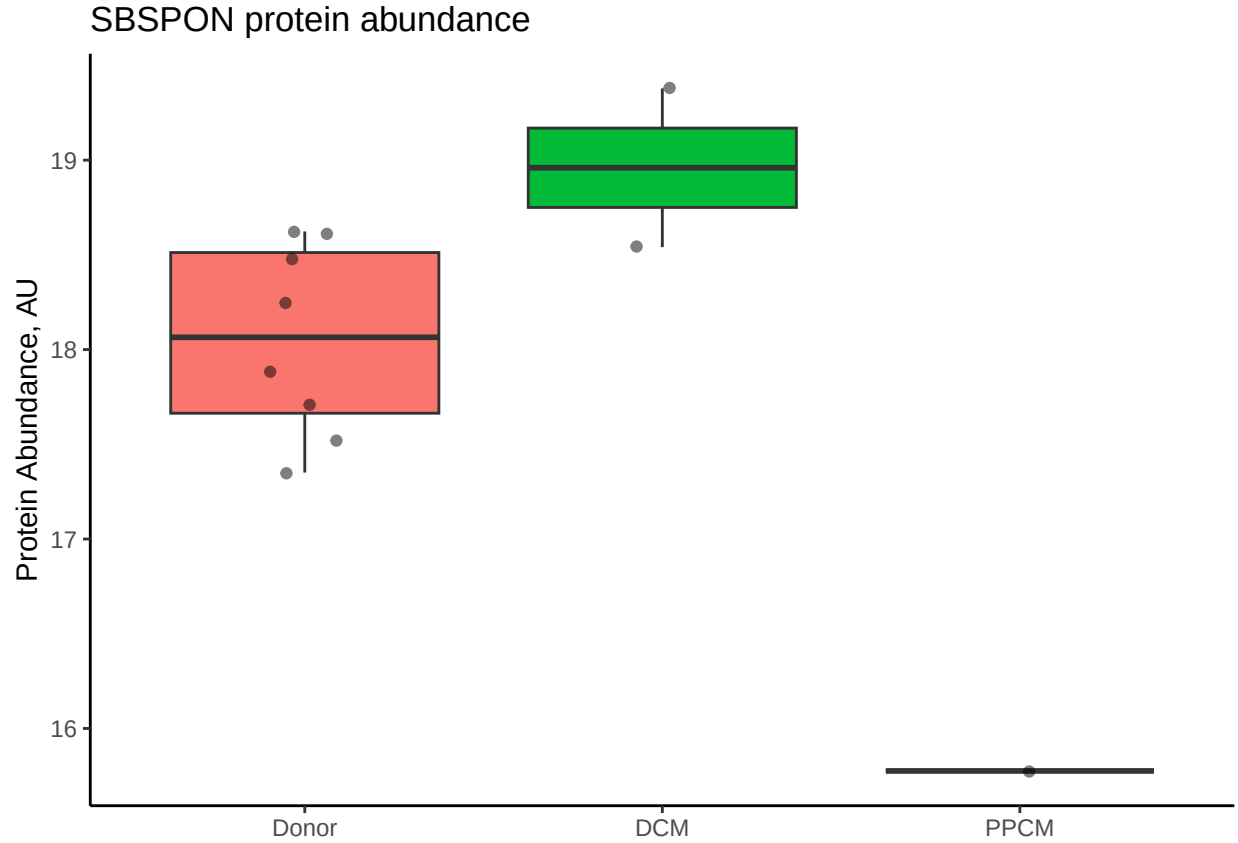
SBSPON is one of 2 proteins that the authors conclude significantly differ in abundance between peripartum cardiomyopathy (PPCM) and dilated cardiomyopathy (DCM). Utilising the data presented in the supplement, we plot the protein abundance across all experimental groups. Note that the data are plotted directly as presented in the supplement, and have not been normalised or (log) transformed as is should be done to quantitatively interpret the differences between groups.

3.1 plot

```

li_2024_abundance_tidy %>%
  filter(gene == "SBSPON") %>%
  ggplot(aes(x = condition, y = abundance, fill = condition)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.5) +
  theme_classic() +
  theme(axis.title.x = element_blank(),
        legend.position = "none") +
  ggtitle("SBSPON protein abundance") +
  labs(y = "Protein Abundance, AU")

```



3.2 count number of valid values

We then produce a table to confirm the number of valid values in each group. This corresponds to the actual sample size used to statistically test for the difference in abundance between the PPCM and DCM groups.

```
li_2024_abundance_tidy %>%
  dplyr::filter(gene == "SBSPON" & !is.na(abundance)) %>%
  group_by(condition) %>%
  summarise(n = n()) %>%
  gt()
```

condition	n
Donor	8
DCM	2
PPCM	1

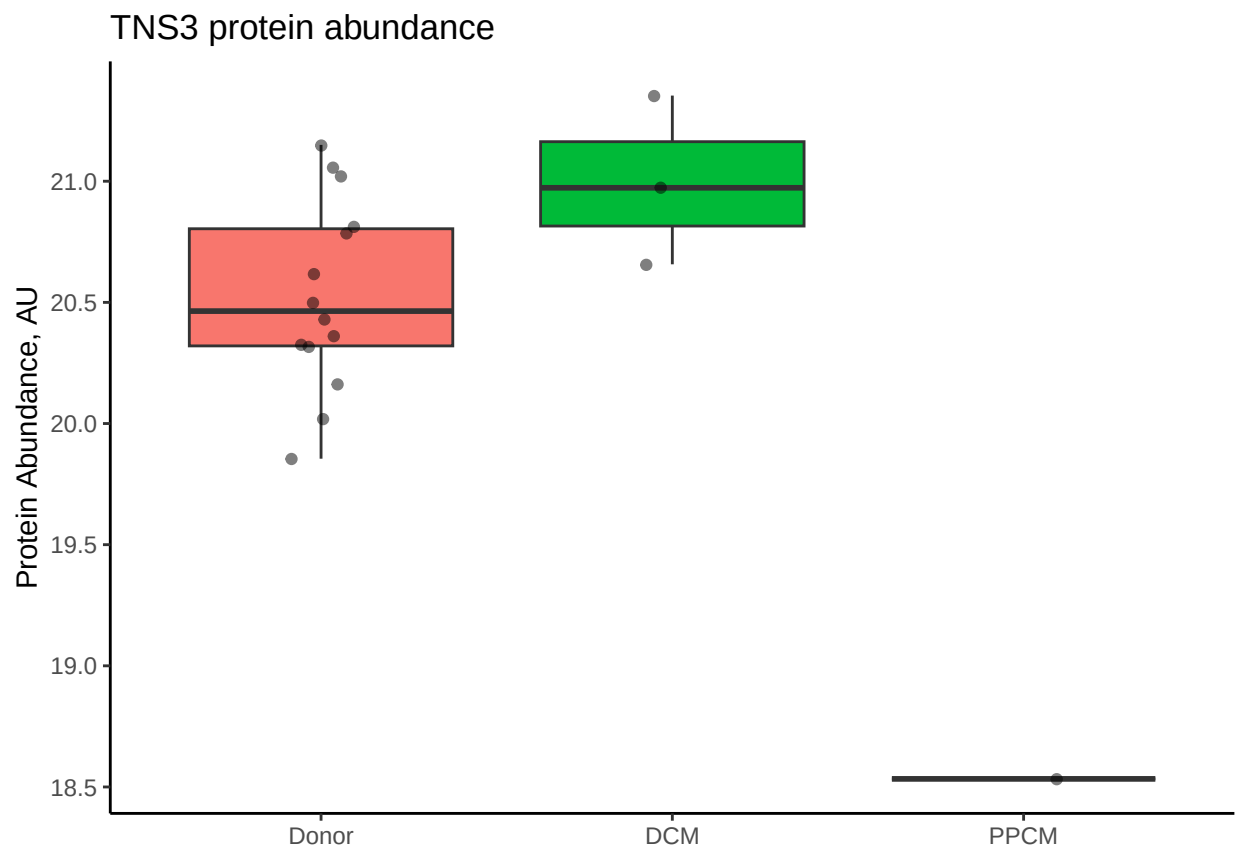
Figure 3H contains 1 datapoint for PPCM and 3 for DCM, meaning again that either the information in Table S1 or Table S2 is incorrect, or a different protein has been plotted in the manuscript.

4 TNS3

TNS3 is the second protein that the authors conclude differs significantly between PPCM and DCM. We similarly plot and produce a table of the number of measured values in each group.

4.1 plot

```
li_2024_abundance_tidy %>%  
  filter(gene == "TNS3") %>%  
  ggplot(aes(x = condition, y = abundance, fill = condition)) +  
  geom_boxplot(outlier.shape = NA) +  
  geom_jitter(width = 0.1, alpha = 0.5) +  
  theme_classic() +  
  theme(axis.title.x = element_blank(),  
        legend.position = "none") +  
  ggtitle("TNS3 protein abundance") +  
  labs(y = "Protein Abundance, AU")
```



4.2 count number of valid values

```
li_2024_abundance_tidy %>%  
  dplyr::filter(gene == "TNS3" & !is.na(abundance)) %>%  
  group_by(condition) %>%  
  summarise(n = n()) %>%  
  gt()
```

condition	n
Donor	14
DCM	3
PPCM	1

The plot of TNS3 in Figure 3H contains 1 data point for PPCM and 5 for DCM, meaning again that either the information in Table S1 or Table S2 is incorrect, or a different protein has been plotted in the manuscript.

```
sessionInfo()
```

```
## R version 4.2.2 (2022-10-31)
## Platform: aarch64-apple-darwin20 (64-bit)
## Running under: macOS Ventura 13.0
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/lib/libRlapack.dylib
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] gt_0.8.0      janitor_2.2.0 readxl_1.4.2  lubridate_1.9.2
## [5] forcats_1.0.0 stringr_1.5.0 dplyr_1.1.1  purrr_1.0.1
## [9] readr_2.1.4   tidyr_1.3.0   tibble_3.2.1 ggplot2_3.4.4
## [13] tidyverse_2.0.0 dbplyr_2.3.1
##
## loaded via a namespace (and not attached):
## [1] highr_0.11      cellranger_1.1.0 pillar_1.9.0    compiler_4.2.2
## [5] tools_4.2.2     digest_0.6.31   timechange_0.2.0 evaluate_0.24.0
## [9] lifecycle_1.0.3 gtable_0.3.3    pkgconfig_2.0.3 rlang_1.1.0
## [13] DBI_1.1.3       cli_3.6.1       rstudioapi_0.14 yaml_2.3.9
## [17] xfun_0.46       fastmap_1.1.1   withr_2.5.0     knitr_1.48
## [21] generics_0.1.3  vctrs_0.6.1     hms_1.1.2       grid_4.2.2
## [25] tidyselect_1.2.0 snakecase_0.11.0 glue_1.6.2      R6_2.5.1
## [29] fansi_1.0.4     rmarkdown_2.21  farver_2.1.1    tzdb_0.3.0
## [33] magrittr_2.0.3  scales_1.2.1    htmltools_0.5.5 ellipsis_0.3.2
## [37] colorspace_2.1-0 labeling_0.4.2   utf8_1.2.3      stringi_1.7.12
## [41] munsell_0.5.0
```